

? Q1. [Answer choice] are physically separate data centers within an Azure region.

Options:

- **A) Availability zones**
- **B) Geographies**
- **C) Region pairs**
- **D) Resource groups**

✓ Answer: A) Availability zones

 **Explanation:**

Availability zones are distinct physical locations within an Azure region, each with its own independent power, cooling, and networking infrastructure. Their primary purpose is to protect applications and data from datacenter-level failures, ensuring high availability and resilience. By distributing VMs across different zones, users can achieve a 99.99% uptime SLA for their critical workloads.

? Q2. Which two components are created in an Azure subscription? (Select 2)

Options:

- **A) Microsoft Entra user accounts**
- **B) Management groups**
- **C) Resource groups**
- **D) Resources**

✓ Answer: C) Resource groups and D) Resources

 **Explanation:**

An Azure subscription serves as a billing and security boundary for all Azure resources. When a subscription is created, it allows you to logically organize related resources into resource groups (management containers) and deploy actual service instances like VMs, databases, or storage accounts as individual resources inside them. Microsoft Entra user accounts exist at the tenant level, not within the subscription itself.

? Q3. What is an Azure Storage account named storage001 an example of?

Options:

- **A) A resource**
- **B) A resource group**

- C) A resource manager
- D) A subscription

✅ **Answer: A) A resource**

 Explanation:

A storage account is an individual, manageable Azure service instance that is deployed and organized inside a resource group. Therefore, it is categorized as a fundamental Azure resource. Resource groups and subscriptions are higher-level management containers, while Azure Resource Manager (ARM) is the management layer responsible for orchestrating resource deployment and operations.

? Q4. Which Azure resource is a software emulation of a physical computer that includes a virtual processor, memory, storage, and networking?

Options:

- A) A container
- B) A function
- C) A virtual machine
- D) An App Service

✅ **Answer: C) A virtual machine**

 Explanation:

Azure Virtual Machines (VMs) provide Infrastructure-as-a-Service (IaaS) by fully emulating a complete physical computer system, giving you full control over the operating system, CPU, RAM, and storage configuration. Containers and Functions are serverless or containerized offerings that abstract away the underlying hardware entirely. The primary purpose of VMs is to host legacy applications and workloads that require a specific operating system environment.

? Q5. What can you use to execute code in a serverless environment?

Options:

- A) Azure Container Instances
- B) Azure Functions
- C) Azure Logic Apps
- D) Azure Virtual Desktop

✅ **Answer: B) Azure Functions**

 Explanation:

Azure Functions is a serverless compute service that allows you to run event-triggered code in response to various events (like HTTP requests, queue messages, or timer schedules) without provisioning or managing any underlying

servers. While Azure Logic Apps uses pre-built workflows, Azure Functions is specifically designed for writing and executing custom code in a serverless model.

? Q6. Which scenario is a use case for a VPN gateway?

Options:

- **A)** Communicating between Azure resources
- **B)** Connecting an on-premises datacenter to an Azure virtual network
- **C)** Filtering outbound network traffic
- **D)** Partitioning a virtual network's address space

✓ Answer: B) Connecting an on-premises datacenter to an Azure virtual network

 **Explanation:**

A VPN Gateway encrypts traffic to create a secure tunnel between your on-premises network and Azure virtual networks. Its primary purpose is to extend your local datacenter to the cloud over the public internet securely. Filtering traffic uses Network Security Groups (NSGs) or Azure Firewall, while partitioning address spaces involves creating subnets within the VNet.

? Q7. What can you use to connect Azure resources, such as Azure SQL databases, to an Azure virtual network?

Options:

- **A)** ExpressRoute
- **B)** Network security groups (NSGs)
- **C)** Peering
- **D)** Service endpoints

✓ Answer: D) Service endpoints

 **Explanation:**

Service endpoints allow you to secure Azure service resources (like Azure SQL Database or Storage accounts) to your Virtual Network by providing a direct, private connection over the Azure backbone network. This effectively bypasses the public internet, significantly enhancing security and routing efficiency for PaaS services.

? Q8. Which two services can you use to establish network connectivity between an on-premises network and Azure resources? (Select 2)

Options:

- **A)** Azure Bastion

- **B) Azure Firewall**
- **C) Azure VPN Gateway**
- **D) ExpressRoute**

 **Answer: C) Azure VPN Gateway and D) ExpressRoute**

 Explanation:

Both VPN Gateway and ExpressRoute are designed to connect on-premises infrastructure to Azure. VPN Gateway is an encrypted tunnel over the internet (site-to-site or point-to-site), while ExpressRoute provides a dedicated, private, high-bandwidth physical connection that completely bypasses the public internet. Bastion and Firewall are security services, not network connectivity services.

? Q9. What can you use to provide Mac and Android users with access to a Windows environment that will run Windows-based applications?

Options:

- **A) Azure Container Instances**
- **B) Azure Functions**
- **C) Azure Logic Apps**
- **D) Azure Virtual Desktop**

 **Answer: D) Azure Virtual Desktop**

 Explanation:

Azure Virtual Desktop (AVD) is a desktop and application virtualization service that allows users to access a full Windows desktop environment from virtually any device, including Mac, Android, iOS, and Linux. Its primary purpose is to provide secure remote work capabilities without the need to manage physical Virtual Desktop Infrastructure (VDI) hardware.

 AZURE STORAGE SERVICES

? Q10. Which storage service should you use to store thousands of files containing text and images?

Options:

- **A) Azure Blob storage**
- **B) Azure Disk Storage**
- **C) Azure Queue Storage**
- **D) Azure Table storage**

 **Answer: A) Azure Blob storage**

 Explanation:

Azure Blob storage is specifically designed to store massive amounts of unstructured data such as text files, images, videos, and documents. Disk storage is for VM disks (block storage), Queue is for messaging and decoupling applications, and Table is for NoSQL key-value data. Its purpose is to serve as a highly scalable object storage solution for big data and media content.

? Q11. Which Azure Blob storage tier stores data offline and offers the lowest storage costs and the highest costs to access data?

Options:

- **A) Archive**
- **B) Cool**
- **C) Hot**
- **D) Cold**

 **Answer: A) Archive**

 Explanation:

The Archive tier is the lowest-cost storage tier, specifically designed for data that is rarely accessed (often requiring several hours to retrieve). It is optimized for long-term retention, backup, and legal compliance scenarios where rapid retrieval isn't necessary. The trade-off is that access costs are significantly higher compared to Hot or Cool tiers.

? Q12. Which two scenarios are common use cases for Azure Blob storage? (Select 2)

Options:

- **A) Hosting ASPX files for a website**
- **B) Mounting a file storage share to be accessed as a virtual drive on multiple virtual machines**
- **C) Serving images or documents directly to a browser**
- **D) Storing data for backup and restore**

 **Answer: C) Serving images or documents directly to a browser and D) Storing data for backup and restore**

 Explanation:

Blob storage excels at serving static web assets (images, CSS, JavaScript, documents) directly to users via HTTP/HTTPS and providing cost-effective, durable storage for backup files. Hosting ASPX files (dynamic web applications) typically requires Azure App Service, and mounting file shares uses Azure Files, not Blob storage.

? Q13. Which Azure Blob storage service tier has the highest storage costs and the fastest access times for reading and writing data?

Options:

- A) Archive
- B) Cool
- C) Hot
- D) Cold

 **Answer: C) Hot**

 **Explanation:**

The Hot storage tier provides the lowest latency for data access and is optimized for frequently accessed data. Because it is designed for maximum performance with millisecond access times, it incurs the highest per-GB storage costs but charges minimal to no retrieval fees. It is ideal for actively used data in production environments.

? Q14. Which Azure Storage service should you use to store unstructured files, such as images, that will be served on webpages?

Options:

- A) Azure Blob storage
- B) Azure Disk Storage
- C) Azure Queue Storage
- D) Azure Table storage

 **Answer: A) Azure Blob storage**

 **Explanation:**

Unstructured file storage, especially for web assets like images, CSS, and videos, is the primary use case for Azure Blob storage. It provides flexible REST APIs and SDKs to serve files directly to end-user browsers. Disk storage is for block-level disks attached to VMs, Queue is for messaging, and Table is for structured NoSQL data.

IDENTITY & ACCESS MANAGEMENT

? Q15. What enables a user to sign in one time and use that credential to access multiple resources and applications from different providers?

Options:

- A) Conditional Access
- B) Device management

- C) Multifactor authentication (MFA)
- D) Single sign-on (SSO)

✅ **Answer: D) Single sign-on (SSO)**

📄 **Explanation:**

Single Sign-On (SSO) is a security feature that allows users to authenticate once and gain access to multiple applications without re-entering their credentials. Its primary purpose is to reduce the number of passwords users must remember, simplify identity management across an enterprise, and improve overall user experience while maintaining security.

? Q16. What can you use to allow a user to manage all the resources in a resource group?

Options:

- A) Azure Key Vault
- B) Azure role-based access control (RBAC)
- C) Resource locks
- D) Resource tags

✅ **Answer: B) Azure role-based access control (RBAC)**

📄 **Explanation:**

Azure RBAC is the authorization system that grants specific permissions (like Contributor, Owner, or Reader) to users, groups, or applications over specific scopes, such as a resource group or subscription. Its purpose is to implement the principle of least privilege by giving users exactly the access they need to manage resources without granting unnecessary permissions.

? Q17. Which type of strategy uses a series of mechanisms to slow the advancement of an attack that aims to gain unauthorized access to data?

Options:

- A) Defense in depth
- B) Distributed denial-of-service (DDoS)
- C) Least privileged access
- D) Perimeter

✅ **Answer: A) Defense in depth**

📄 **Explanation:**

Defense in depth is a layered security approach that uses multiple protective controls (like firewalls, encryption, authentication, and monitoring) across different layers (data, network, application, and physical). The purpose is to ensure that if

one layer is breached or compromised, subsequent layers will still protect the data, slowing the advancement of attacks.

? Q18. What must you use to create hybrid identities in a Microsoft Entra tenant?

Options:

- **A) Microsoft Entra Connect Sync**
- **B) Azure Key Vault**
- **C) Azure Resource Manager (ARM)**
- **D) Conditional Access**

✅ Answer: A) Microsoft Entra Connect Sync

📄 Explanation:

To create hybrid identities (where users exist in both on-premises Active Directory and the cloud), you must use Microsoft Entra Connect Sync (formerly Azure AD Connect). Its purpose is to synchronize user accounts, passwords, and groups between on-premises AD and Microsoft Entra ID so users can use a single identity to access both on-premises and cloud environments.

💰 AZURE COST MANAGEMENT

? Q19. Which two factors affect Azure costs? (Select 2)

Options:

- **A) Availability zone selection**
- **B) Date and time of use**
- **C) Resource location**
- **D) Resource usage**

✅ Answer: C) Resource location and D) Resource usage

📄 Explanation:

Azure pricing heavily depends on the geographic region (resource location) due to differences in datacenter operating costs, electricity, and local market conditions. Additionally, the actual quantity of resources consumed (resource usage, e.g., CPU time, storage GB, data transfer) directly affects billing. Date and time of use generally does not affect base rates, as Azure uses a fixed pay-as-you-go pricing model per region.

? Q20. Which are two common scenarios for using resource tags? (Select 2)

Options:

- **A)** Associating costs with different environments
- **B)** Categorizing costs by department
- **C)** Identifying lower cost regions
- **D)** Resizing underutilized virtual machines

 **Answer: A) Associating costs with different environments and B) Categorizing costs by department**

 **Explanation:**

Resource tags are metadata key-value pairs used to logically organize Azure resources. Their main purpose is cost attribution, allowing administrators to generate detailed billing reports broken down by cost centers (departments) or specific environments (dev, test, production). They do not affect pricing, location, or resource performance.

 **Q21. Which two features are available by using Azure Cost Management + Billing? (Select 2)**

Options:

- **A)** Create and manage budgets
- **B)** Estimate TCO before deploying
- **C)** Generate historical reports and forecast future usage
- **D)** Provide discounted prices when you pay in advance

 **Answer: A) Create and manage budgets and C) Generate historical reports and forecast future usage**

 **Explanation:**

Azure Cost Management helps you monitor spending by setting strict budget alerts (with threshold-based notifications) and analyzing historical consumption data to forecast future usage trends. Estimated TCO is done via the Azure Pricing Calculator, and upfront discounts are applied via Azure Reservations, not directly by Cost Management.

 **Q22. You have an Azure virtual machine accessed only between 9:00 and 17:00 each day. How to minimize costs but preserve hard disks and data?**

Options:

- **A)** Deallocate the VM when not needed
- **B)** Delete the VM when not needed
- **C)** Implement Privileged Identity Management
- **D)** Resize the VM to a smaller size

 **Answer: A) Deallocate the virtual machine when it is not needed**

 Explanation:

Stopping (deallocating) a VM releases the compute resources (CPU and RAM) so you are not billed for compute time, while the associated managed disks and data remain intact and are stored at a much lower storage cost. Deleting the VM would also remove the disks, and resizing doesn't stop billing for compute usage.

? Q23. You need to associate the costs of resources to different groups without changing the location. What should you use?

Options:

- A) Administrative units
- B) Resource groups
- C) Resource tags
- D) Subscriptions

 **Answer: C) Resource tags**

 Explanation:

Resource tags are key-value pairs that act as metadata to categorize billing data. They allow you to track and allocate costs based on department, environment, or project without physically moving the resource to a different region, group, or subscription. This provides flexible cost attribution without impacting the resource's location or functionality.

? Q24. Which two factors affect Azure costs? (Select 2)

Options:

- A) Outbound data transfers
- B) The location of a resource
- C) The number of resource groups in a subscription
- D) The use of ARM templates

 **Answer: A) Outbound data transfers and B) The location of a resource**

 Explanation:

Moving data out of Azure (outbound transfers) incurs bandwidth costs depending on the volume and destination region, and geographic regions have different standard pricing structures for compute and storage services. Resource groups and ARM templates are free management layers and have no impact on billing costs.

? Q25. What can be applied to a resource to prevent accidental deletion?

Options:

- **A) A resource lock**
- **B) A resource tag**
- **C) A policy**
- **D) An Azure Reservation**

✅ Answer: A) A resource lock

📖 Explanation:

An Azure Resource Lock (either CanNotDelete or ReadOnly) strictly prevents users from modifying or deleting locked resources. Its purpose is to safeguard critical assets, such as network gateways, production databases, or important configuration files, against accidental operational mistakes or intentional deletions.

? Q26. What can you use to ensure that new and existing Azure resources stay in compliance with corporate standards?

Options:

- **A) Azure Advisor**
- **B) Azure Policy**
- **C) Resource locks**
- **D) Resource tags**

✅ Answer: B) Azure Policy

📖 Explanation:

Azure Policy is a governance service that uses rules (JSON-based policy definitions) to evaluate, audit, and enforce compliance across all resources. It can automatically deny non-compliant deployments, correct misconfigurations, or audit existing resources to ensure corporate standards are maintained continuously.

? Q27. You need to recommend a solution for VM deployments that enforces company standards on VMs. What should you include?

Options:

- **A) Microsoft Intune**
- **B) Azure Cost Management**
- **C) Azure Lock**
- **D) Azure Policy**

✅ Answer: D) Azure Policy

 Explanation:

Azure Policy is the right tool to enforce company rules on resource configuration (e.g., ensuring all VMs use specific SKUs, have required tags, or follow network security rules). Intune is for managing client devices, not Azure infrastructure, while Azure Locks prevent deletion but don't enforce configuration standards.

? Q28. Resource locks can be applied to a subscription, resource group, or [Target_A].

Options:

- A) Azure region
- B) Individual resource
- C) Management group
- D) PowerShell script

 **Answer: B) Individual resource**

 Explanation:

Resource locks are scoped hierarchically, meaning they can be applied at the subscription level (affecting all resources within), resource group level (affecting all resources in the group), or individual resource level (affecting just that specific resource). The lock applies to the specific resource and all its child resources. They cannot be applied to a script or an Azure region directly.

? Q29. What can you use to define the resources you want to provision in a declarative JSON format?

Options:

- A) Azure CLI
- B) Azure PowerShell
- C) Azure Repos
- D) Azure Resource Manager (ARM) templates

 **Answer: D) Azure Resource Manager (ARM) templates**

 Explanation:

ARM templates are Infrastructure-as-Code (IaC) files written in JSON format that declaratively define the exact Azure resources, dependencies, and configuration needed for a deployment. Azure Repos is for storing code, while CLI and PowerShell are imperative scripting tools for interactive management.

? Q30. Which management layer accepts requests from any Azure tool or API and enables you to create/update/delete resources?

Options:

- A) Azure CLI
- B) Azure management groups
- C) Azure Resource Manager (ARM)
- D) Azure Sphere

✅ **Answer: C) Azure Resource Manager (ARM)**

📄 **Explanation:**

The Azure Resource Manager (ARM) is the true orchestration layer that sits behind the scenes of all Azure tools and APIs. It authenticates requests, validates constraints and quotas, and orchestrates the provisioning of resources via REST APIs. Azure CLI and PowerShell are just client tools that talk to this ARM layer.

? **Q31. Which two tools are accessible via Azure Cloud Shell to manage an Azure environment? (Select 2)**

Options:

- A) Azure CLI
- B) Azure PowerShell
- C) Azure Repos
- D) Azure Resource Manager (ARM) templates

✅ **Answer: A) Azure CLI and B) Azure PowerShell**

📄 **Explanation:**

Azure Cloud Shell is a browser-based interactive shell that automatically comes configured with both Azure CLI and Azure PowerShell available out of the box. While it can store ARM templates in the persistent storage, the interactive command-line tools available for management are CLI and PowerShell.

? **Q32. What can you use to manage servers across third party cloud platforms and on-premises environments?**

Options:

- A) Azure Arc
- B) Azure CLI
- C) Azure Monitor
- D) Azure PowerShell

✅ **Answer: A) Azure Arc**

📄 **Explanation:**

Azure Arc allows you to extend Azure's management, governance, and security controls to servers that live outside of Azure, including other cloud platforms (AWS,

GCP) and on-premises datacenter servers. Its purpose is to provide a single pane of glass for all your infrastructure, regardless of where it is hosted.

? Q33. You have on-premises servers. Need to centralize management without migrating them to Azure. What should you use?

Options:

- A) Azure Advisor
- B) Azure Arc
- C) Azure Monitor
- D) Azure Service Health

✅ Answer: B) Azure Arc

📄 Explanation:

Azure Arc is the specific tool to centrally manage servers physically located on-premises or in other clouds. Azure Advisor only recommends best practices for native Azure resources, while Service Health monitors Azure's actual uptime and service status.

? Q34. You need to create a custom solution that uses thresholds to trigger autoscaling functionality to scale an app up or down. What should you include?

Options:

- A) Application Insights
- B) Azure Advisor
- C) Azure Monitor
- D) Azure Service Health

✅ Answer: C) Azure Monitor

📄 Explanation:

Azure Monitor collects telemetry, metrics, and logs (like CPU percentage, memory usage, or request queue length). Its purpose is to feed these metrics into an Autoscale rule so that when a pre-defined threshold is hit, the application scales automatically to meet demand or reduce costs.

? Q35. What can you use to automatically detect performance anomalies for web apps?

Options:

- A) Azure Advisor
- B) Azure Application Insights

- C) Azure Cognitive Services
- D) Azure DevOps

✅ **Answer: B) Azure Application Insights**

📄 **Explanation:**

Application Insights is a dedicated feature of Azure Monitor specifically designed for Application Performance Management (APM). It automatically detects abnormal spikes in response times, failure rates, and server requests, alerting developers to performance anomalies before they impact users.

☁️ CLOUD COMPUTING CONCEPTS

? **Q36. What are two characteristics of a consumption-based model? (Select 2)**

Options:

- A) High capital expenditures
- B) No upfront costs
- C) Requires purchase of physical infrastructure
- D) Ability to stop paying for unused resources

✅ **Answer: B) No upfront costs and D) Ability to stop paying for unused resources**

📄 **Explanation:**

The cloud consumption model (pay-as-you-go) shifts costs from Capital Expenditure (CapEx) to Operating Expenditure (OpEx). Its main purposes are to eliminate heavy upfront hardware investments and provide the flexibility to completely deprovision resources to stop billing immediately when they are no longer needed.

? **Q37. Which two characteristics are common advantages of cloud computing? (Select 2)**

Options:

- A) Elimination of horizontal scaling
- B) Geo-distribution
- C) High availability
- D) Physical access to servers

✅ **Answer: B) Geo-distribution and C) High availability**

📄 **Explanation:**

Cloud computing allows you to distribute applications to datacenters globally (geo-

distribution) for local performance and low latency, and design systems with no single point of failure (high availability). The cloud provider manages physical hardware access, and horizontal scaling is heavily utilized (not eliminated) in cloud architectures.

? Q38. Which cloud deployment model are you using if you have servers on-site and migrate a few servers to the cloud?

Options:

- **A) Hybrid cloud**
- **B) Private cloud**
- **C) Public cloud**

✓ Answer: A) Hybrid cloud

📝 Explanation:

A hybrid cloud mixes on-premises (private) infrastructure with public cloud resources. Its purpose is to allow organizations to keep sensitive data on-premises while leveraging public cloud's compute scalability for burst workloads, all connected through secure VPN or ExpressRoute connections.

? Q39. Deploying and configuring cloud-based resources quickly as business requirements change is called:

Options:

- **A) Agility**
- **B) Elasticity**
- **C) High availability**
- **D) Scalability**

✓ Answer: A) Agility

📝 Explanation:

Agility is the ability to spin up, configure, and update resources in minutes rather than weeks. The purpose is to respond rapidly to market shifts and changing business requirements. While Elasticity handles automatic scaling based on demand, Agility specifically refers to the speed and flexibility of the deployment process itself.

? Q40. Increasing compute capacity for an app by adding instances of resources such as VMs is called:

Options:

- **A) Disaster recovery**
- **B) High availability**

- C) Horizontal scaling
- D) Vertical scaling

✓ **Answer: C) Horizontal scaling**

 **Explanation:**

Horizontal scaling (also called scaling out) means distributing the workload across multiple compute instances (adding more VMs or app instances) to handle increased traffic. Its purpose is to achieve massive scalability that is much higher than what a single massive VM (vertical scaling) could provide, with better fault tolerance.

? Q41. What is high availability in a public cloud environment dependent on?

Options:

- A) Capital expenditures
- B) Backup retention limits
- C) The service-level agreement (SLA)
- D) Vertical scalability

✓ **Answer: C) The service-level agreement (SLA)**

 **Explanation:**

High availability uptime percentages (like 99.9% or 99.99%) are legally contractually guaranteed by the cloud provider through the Service Level Agreement (SLA). Different service tiers offer different SLAs, and these directly define the guaranteed availability of your application and the compensation you receive if the guarantee is not met.

? Q42. Increasing the capacity of an application by adding additional virtual machines is called:

Options:

- A) Agility
- B) High availability
- C) Horizontal scaling
- D) Vertical scaling

✓ **Answer: C) Horizontal scaling**

 **Explanation:**

Adding more VMs to an existing pool to distribute the workload is the textbook definition of horizontal scaling (scale out). This approach is vital for cloud-based web applications that experience fluctuating traffic demands and need to handle large volumes of concurrent users efficiently.

CLOUD SERVICE MODELS

? Q43. Which cloud service model provides you with the most control?

Options:

- **A)** Infrastructure as a Service (IaaS)
- **B)** Platform as a Service (PaaS)
- **C)** Software as a Service (SaaS)

✓ Answer: A) Infrastructure as a Service (IaaS)

 **Explanation:**

IaaS gives the customer the most control by providing virtualized hardware (computing, networking, storage) and allowing them to manage the operating system, middleware, and applications. The purpose is for legacy lift-and-shift scenarios where you need to manage the entire OS stack or have specific configuration requirements.

? Q44. In which cloud service model is the customer responsible for managing the operating system?

Options:

- **A)** Infrastructure as a Service (IaaS)
- **B)** Platform as a Service (PaaS)
- **C)** Software as a Service (SaaS)

✓ Answer: A) Infrastructure as a Service (IaaS)

 **Explanation:**

In IaaS, the cloud provider manages the physical hardware, but the customer is entirely responsible for patching, configuring, and managing the operating system on their virtual machines. In PaaS and SaaS, the provider manages the operating system entirely on behalf of the customer, abstracting away the underlying infrastructure.

? Q45. What is the customer responsible for in a Software as a Service (SaaS) model?

Options:

- **A)** Data and access
- **B)** Storage
- **C)** Runtime

- D) Virtual machines

 **Answer: A) Data and access**

 **Explanation:**

In the SaaS model, the provider manages everything from the physical infrastructure up to the application code. The customer is only responsible for managing their own business data stored in the application and configuring user access and permissions via the application's interface.

? Q46. Your organization is building a custom application and you need to focus on coding rather than servers. Which model?

Options:

- A) IaaS
- B) PaaS
- C) SaaS

 **Answer: B) Platform as a Service (PaaS)**

 **Explanation:**

PaaS (like Azure App Services) abstracts the underlying server management, OS patching, and runtime environment completely. Its purpose is to let developers focus purely on writing application code and deploying it, without worrying about the host infrastructure or OS configuration.

? Q47. Which cloud service model is used by Microsoft 365?

Options:

- A) IaaS
- B) PaaS
- C) SaaS

 **Answer: C) Software as a Service (SaaS)**

 **Explanation:**

Microsoft 365 is a prime example of SaaS. It is a fully pre-built application running on Microsoft's cloud infrastructure. The purpose is to allow users to connect and use office tools (Word, Excel, Outlook, Teams) via the internet instantly without needing to install, patch, or manage any software or servers.

? Q48. Which cloud service model is used by Azure SQL Database?

Options:

- A) IaaS
- B) PaaS

- C) SaaS

✅ **Answer: B) Platform as a Service (PaaS)**

📄 **Explanation:**

Azure SQL Database is a fully managed PaaS database engine. While you write SQL queries and manage database schemas, Microsoft automatically handles backups, updates, scaling, and patching of the underlying servers. Its purpose is to reduce administrative overhead for database administration.

? **Q49. What uses the Infrastructure as a Service (IaaS) cloud service model?**

Options:

- A) Azure App Services
- B) Azure Cosmos DB
- C) Azure virtual machines
- D) Microsoft 365

✅ **Answer: C) Azure virtual machines**

📄 **Explanation:**

Azure Virtual Machines are the standard IaaS offering, where the user has complete control over the operating system and configurations. App Services and Cosmos DB are PaaS offerings, and Microsoft 365 is SaaS. The purpose of IaaS VMs is to run workloads that require specific OS configurations or legacy applications.

? **Q50. What are Azure Service Models?**

Options:

- A) IaaS, PaaS, SaaS
- B) Public, Private, Hybrid
- C) Compute, Storage, Networking
- D) Free, Basic, Standard, Premium

✅ **Answer: A) IaaS, PaaS, SaaS**

📄 **Explanation:**

Azure Service Models define the level of abstraction and control provided to customers:

- **IaaS (Infrastructure as a Service)** → Full control over OS, storage, and network (VMs)
- **PaaS (Platform as a Service)** → Platform-level services without managing servers (App Service, SQL Database)

- **SaaS (Software as a Service)** → Fully managed applications (Microsoft 365, Dynamics 365)
Control decreases from IaaS → PaaS → SaaS, while management overhead decreases in the same direction.

COMPLETE REFERENCE SUMMARY

Azure Service Models Comparison

Feature	IaaS	PaaS	SaaS
Control	Highest	Medium	Lowest
Management	Customer manages OS, apps	Provider manages OS, runtime	Provider manages everything
Examples	VMs, Storage, VNet	App Service, SQL DB	Office 365, Dynamics
Use Case	Lift & Shift	Custom Development	Ready-to-use Apps

Key Azure Services Quick Reference

Service	Category	Purpose
Azure Virtual Machines	IaaS	Full OS control, legacy workloads
Azure App Service	PaaS	Web app hosting, API management
Azure Functions	Serverless	Event-driven code execution
Azure SQL Database	PaaS	Managed relational database
Azure Blob Storage	Storage	Unstructured data, media files
Azure AD / Entra ID	Identity	Authentication, SSO, MFA
Azure VPN Gateway	Networking	On-premises connection via internet

Service	Category	Purpose
ExpressRoute	Networking	Private dedicated connection
Azure Virtual Desktop	Virtualization	Remote Windows desktop access
Azure Arc	Management	Multi-cloud & hybrid management
Azure Policy	Governance	Compliance enforcement
Azure Monitor	Monitoring	Metrics, logs, alerts
Application Insights	Monitoring	APM, performance monitoring
Azure Key Vault	Security	Secrets, keys, certificates management
Azure Firewall	Security	Centralized network traffic filtering
Azure Bastion	Security	Secure RDP/SSH access without public IP
Azure Site Recovery	DR	Disaster recovery, failover

Cost Management Strategies

Strategy	Description	Savings
Resource Tags	Categorize costs by department/environment	Better visibility
Deallocate VMs	Stop billing when not in use	Up to 70%
Reserved Instances	Pre-pay for 1-3 years	Up to 72%
Auto-scaling	Scale down during low usage	Up to 40%
Spot VMs	Use spare capacity at low cost	Up to 90%

Security Best Practices

Practice	Purpose
RBAC	Implement least privilege access
NSG	Filter network traffic at subnet/VM level
Azure Firewall	Centralized network protection
Private Endpoint	Private access to Azure services
Key Vault	Secure secrets, keys, and certificates
Defense in Depth	Multi-layer security approach
Zero Trust	Never trust, always verify
Azure Policy	Enforce security standards

Hybrid Connectivity Options

Service	Speed	Security	Use Case
VPN Gateway	Up to 1.25 Gbps	Encrypted over internet	S2S, P2S connections
ExpressRoute	Up to 10 Gbps	Private dedicated circuit	Enterprise production workloads

Scaling Strategies

Strategy	Description	Best For
Horizontal Scaling	Add more instances (scale out)	Web apps, microservices
Vertical Scaling	Increase size of existing instance (scale up)	Databases, heavy workloads
Auto-scaling	Automatic scaling based on metrics	Variable workloads

Strategy	Description	Best For
Availability Set	Protects against hardware failures	Production VMs
Scale Set	Group of identical VMs with auto-scaling	Large-scale applications

Exam Quick Notes

Topic	Key Points
Availability Zones	Physical datacenters within same region → 99.99% SLA
Region Pairs	Two regions for disaster recovery (same geography)
Resource Groups	Logical container for resources (free tier)
Management Groups	Hierarchical organization of subscriptions
Azure Policy	Enforces compliance, can deny/deploy resources
RBAC	Role-based access control (Owner, Contributor, Reader)
Resource Locks	Prevent accidental deletion (CanNotDelete, ReadOnly)
ARM Templates	Declarative JSON for infrastructure-as-code

What are the core services in Azure DevOps?

- **Azure Repos** → Source code management (Git/TFVC)
- **Azure Pipelines** → CI/CD automation
- **Azure Boards** → Work tracking
- **Azure Artifacts** → Package management
- **Azure Test Plans** → Testing tools

One-liner:

Azure DevOps provides Repos, Pipelines, Boards, Artifacts, and Test Plans.

What is CI/CD?

- **CI (Continuous Integration):**
 - Code integration with automated builds & tests.
- **CD (Continuous Delivery/Deployment):**
 - Automated release to environments.

One-liner:

CI/CD automates build, test, and deployment processes.

YAML vs Classic Pipeline

- **YAML:**
 - Code-based pipeline (stored in repo)
 - Version-controlled
- **Classic:**
 - UI-based
 - Easier but less flexible

One-liner:

YAML is code-driven; Classic is UI-driven.

What is a Self-hosted Agent?

- Agent installed on your own server.
- Used when custom environment or tools required.
- More control than Microsoft-hosted agents.

One-liner:

Self-hosted agent runs pipelines on your own infrastructure.

What is Microsoft-hosted Agent?

- Managed by Microsoft.
- Pre-configured environments.
- No maintenance needed.

One-liner:

Microsoft-hosted agents are ready-to-use build environments.

What are stages, jobs, and tasks?

- **Stage:** Logical group (Build, Deploy)
- **Job:** Runs on an agent
- **Task:** Individual step (script, tool)

One-liner:

Stage → Job → Task defines pipeline execution hierarchy.

What are variables in Azure DevOps?

- Store values used in pipelines.
- Types:
 - Pipeline variables
 - Variable groups
 - Secret variables

One-liner:

Variables store reusable configuration values in pipelines.

How do you secure secrets in pipelines?

- Use Azure Key Vault.
- Store secrets as secure variables.
- Restrict access via RBAC.
- Never hardcode credentials.

One-liner:

I use Key Vault and secure variables to protect secrets.

What is Service Connection?

- Secure connection to external services.
- Used for Azure, Docker, GitHub, etc.

One-liner:

Service connection links Azure DevOps to external resources securely.

Pipeline is failing – how do you troubleshoot?

- Check pipeline logs (first step always).
- Identify failing task/stage.
- Verify:
 - Variables
 - Service connections
 - Agent compatibility
- Re-run with debug (system.debug=true).
- Test locally if possible.

One-liner:

I isolate the failure using logs, validate configs, and debug step-by-step.

Deployment failed in production – what will you do?

- Check logs and identify root cause.
- Rollback to previous stable version.
- Fix issue in lower environment.
- Redeploy after validation.

One-liner:

I ensure quick rollback and fix issues before redeploying.

How do you handle zero-downtime deployment?

- Use:
 - Blue-Green deployment
 - Rolling updates
- Load balancer switches traffic.
- Monitor health before full rollout.

One-liner:

I use blue-green or rolling strategies to ensure zero downtime.

Pipeline is slow – how to optimize?

- Enable caching.
- Use parallel jobs.
- Remove unnecessary tasks.
- Use lightweight agents.

One-liner:

I optimize pipelines using caching, parallelism, and task reduction.

How do you implement approvals in pipeline?

- Use environment approvals.
- Add manual validation step.
- Restrict based on roles.

One-liner:

I use environment-based approvals for controlled deployments.

Agent not picking job – why?

- Agent offline
- Capability mismatch
- Pool misconfiguration

One-liner:

Check agent status, capabilities, and pool configuration.

Pipeline triggered but not running

- Trigger misconfiguration
- Branch filter issue

- YAML syntax error

One-liner:

Verify triggers, branch filters, and YAML correctness.

Permission denied error

- Check RBAC roles
- Validate service connection
- Ensure access to resources

One-liner:

Fix access using proper roles and service connection permissions.

How do you design reusable pipelines?

- Use YAML templates (template: keyword).
- Create centralized pipeline repo.
- Pass parameters dynamically.
- Promote DRY (Don't Repeat Yourself).

One-liner:

I use YAML templates and parameters to build reusable, scalable pipelines.

2. How do you implement pipeline as code best practices?

- Store YAML in Azure Repos.
- Use version control and PR reviews.
- Use modular templates.
- Maintain naming conventions.

One-liner:

I treat pipelines as code with versioning, templates, and reviews.

How do you manage multi-branch pipelines?

- Use branch filters (include/exclude).
- Separate pipelines for:
 - feature
 - develop
 - main
- Apply PR validation builds.

One-liner:

I use branch-based triggers and PR validation for multi-branch strategy.

How would you migrate on-prem CI/CD to Azure DevOps?

- Analyze current tools and workflows.
- Move code to Azure Repos.
- Recreate pipelines in YAML.
- Set up agents (self-hosted if needed).
- Gradual migration (not big bang).

One-liner:

I migrate incrementally with minimal disruption using YAML pipelines.

How do you handle secrets leakage incident?

- Rotate all credentials immediately.
- Revoke compromised access.
- Audit logs.
- Strengthen policies.

One-liner:

I respond with rotation, revocation, and audit-based prevention.

What challenges did you face in your project?

- Pipeline failures due to environment mismatch.
- Secret management issues.
- Deployment downtime.
- Slow pipeline performance.

How you solved:

- Used variable groups + Key Vault.
- Implemented blue-green deployment.
- Optimized pipelines with caching.

One-liner:

I solved pipeline, security, and deployment issues using automation and best practices.

Describe your CI/CD pipeline flow

- Code pushed to repo (Azure Repos).
- CI:
 - Build + test
- Artifact stored
- CD:
 - Deploy Dev → QA → Prod
- Approval before production

One-liner:

My pipeline ensures automated build, test, and controlled deployment.

Difference between Continuous Delivery vs Continuous Deployment

- **Delivery:** Manual approval before prod.
- **Deployment:** Fully automated to production.

One-liner:

Delivery needs approval; Deployment is fully automated.

Why YAML pipelines over Classic?

- Version-controlled
- Reusable
- Scalable
- Supports templates

One-liner:

YAML pipelines are flexible, versioned, and scalable.

How do you integrate Terraform in pipeline?

- Install Terraform in pipeline.
- Run:
 - terraform init
 - terraform plan
 - terraform apply
- Store state securely.

One-liner:

I automate infrastructure provisioning using Terraform in pipelines.

What are Azure Service Models?

Answer:

- IaaS → Infrastructure (VMs, Network)
- PaaS → Platform (App Service, DB)
- SaaS → Software (Office 365)
- Control decreases from IaaS → SaaS

One-liner:

Azure service models define levels of control: IaaS (full), PaaS (partial), SaaS (minimal).

What is Azure Active Directory?**Answer :**

- Identity management service
- Provides authentication & authorization
- Supports SSO
- Integrates with apps

One-liner:

Azure AD manages user identity and access securely.

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RBAC controls who can access Azure resources.

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- Protects against failures

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Availability Set ensures VM uptime during failures and maintenance.

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- Manages group of identical VMs
- Supports auto-scaling
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Scale Set automatically manages and scales multiple VMs.

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Application Gateway is a Layer 7 load balancer with advanced routing.

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Answer :

- Load Balancer → Layer 4 (TCP/UDP)
- App Gateway → Layer 7 (HTTP/HTTPS)
- LB → basic traffic distribution
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Load Balancer works at L4, App Gateway works at L7 with intelligent routing.

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- Disaster recovery solution

- Replicates workloads
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Answer :

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Bastion provides secure VM access without exposing public IP.

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Answer :

- Global load balancer
- Layer 7 routing
- Low latency routing
- CDN + WAF integration

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Front Door routes global traffic with low latency and high availability.

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- Service Endpoint → Public IP
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- Service Endpoint easier setup

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Answer :

- Predefined cloud environment setup
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Landing Zone is a secure, scalable foundation for Azure environments.

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Zero Trust means verify everything before granting access.

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AZ is local redundancy, Region Pair is global disaster recovery.

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- No internet exposure
- Used by enterprises

One-liner:

ExpressRoute provides private, high-speed connection to Azure.

How do you design a highly available architecture in Azure?**Answer :**

- Use **Availability Zones**
- Deploy multiple VMs / instances

- Add **Load Balancer / Application Gateway**
- Use **Auto Scaling**
- Implement backup & disaster recovery

One-liner:

High availability is achieved using redundancy, load balancing, and failover mechanisms.

[How do you design a scalable architecture?](#)

Answer :

- Use **VM Scale Sets / App Service scaling**
- Enable auto-scaling based on metrics
- Use stateless architecture
- Add caching (Redis)
- Use CDN

One-liner:

Scalability is achieved by auto-scaling and stateless architecture design.

[How do you secure Azure resources?](#)

Answer :

- Use **RBAC (least privilege)**
- Enable **NSG + Firewall**
- Use **Private Endpoints**
- Store secrets in **Key Vault**
- Enable monitoring & alerts

One-liner:

Security is ensured using RBAC, network isolation, and secret management.

[How do you optimize cost in Azure?](#)

Answer :

- Use **Reserved Instances**
- Enable auto-scaling
- Shut down unused VMs
- Use **Spot VMs**
- Monitor via cost management

One-liner:

Cost optimization is done using scaling, reservations, and monitoring usage.

[How do you connect on-premises to Azure?](#)

Answer :

- Use **VPN Gateway** (quick setup)
- Use **ExpressRoute** (private connection)
- Configure VNet integration
- Secure with NSG & firewall

One-liner:

Hybrid connectivity is achieved via VPN or ExpressRoute.

[How do you handle disaster recovery?](#)

Answer :

- Use **Azure Site Recovery**
- Replicate to another region
- Use backups
- Test failover regularly
- Define RTO & RPO

One-liner:

Disaster recovery ensures business continuity via replication and failover.

Difference: Azure VM vs App Service

Answer :

- VM → Full control
- App Service → Managed platform
- VM → OS level access
- App Service → No OS management

One-liner:

VM gives full control, App Service is fully managed

What is fault domain vs update domain?

Answer :

- Fault domain → Hardware failure isolation
- Update domain → Maintenance isolation
- Used in Availability Set
- Prevents downtime

One-liner:

Fault domain = hardware isolation, update domain = maintenance isolation.

How do you reduce latency globally?

Answer :

- Use **Azure Front Door**
- Use **CDN for static content**
- Deploy apps in multiple regions
- Use Traffic Manager

One-liner:

Global latency is reduced using CDN and global load balancing.

How do you implement Zero Downtime Deployment?

Answer :

- Use **deployment slots (App Service)**
- Blue-Green deployment
- Rolling updates
- Health checks before swap

One-liner:

Zero downtime is achieved using blue-green or rolling deployments.

How do you enforce governance in Azure?

Answer :

- Use **Azure Policy**
- Use **Blueprints**
- Tagging strategy
- Role-based access

One-liner:

Governance is enforced using policies, roles, and standards.

Hub-Spoke Architecture in Azure?

Answer :

- Hub → central services (firewall, VPN)
- Spoke → application VNets
- Improves security
- Scalable design

One-liner:

Hub-Spoke centralizes networking and improves security.

What is Azure Data Factory?

Answer :

- ETL/ELT service
- Data integration
- Supports pipelines
- Connects multiple sources

One-liner:

Data Factory is used for data integration and transformation.

How do you choose between Azure services?

Answer :

- Based on workload
- Cost vs performance
- Scalability needs
- Management overhead
- Security requirements

One-liner:

Service selection depends on workload, cost, and scalability needs.

What is the biggest challenge in cloud?

Answer :

- Cost management
- Security
- Complexity
- Skill gap
- Governance

One-liner:

Managing cost, security, and complexity is the biggest challenge.

What is your approach to troubleshooting in Azure?

Answer :

- Check logs (Monitor)
- Validate configuration
- Check network rules
- Use diagnostics
- Reproduce issue

One-liner:

Troubleshooting involves logs, configuration checks, and diagnostics.

How do you handle production issues?

Answer :

- Identify root cause
- Check monitoring alerts
- Apply quick fix
- Communicate with team
- Prevent future issues

One-liner:

Handle production issues by fixing fast and preventing recurrence.

How do you design a secure network architecture in Azure?

Answer :

- Use **VNet with subnets (segmentation)**
- Apply **NSG rules**
- Use **Azure Firewall** for centralized control
- Implement **Private Endpoints**

- Enable DDoS protection

One-liner:

Secure architecture uses network isolation, firewalls, and private access.

. Difference: Azure Front Door vs Traffic Manager

Answer :

- Front Door → Layer 7 (HTTP/HTTPS)
- Traffic Manager → DNS-based
- Front Door → real-time routing
- Traffic Manager → DNS resolution-based

One-liner:

Front Door is L7 routing, Traffic Manager is DNS-based routing.

How do you handle secrets in CI/CD?

Answer:

- Use **Azure Key Vault**
- Integrate with pipeline
- Avoid hardcoding
- Use Managed Identity
- Rotate secrets

One-liner:

Secrets are managed securely using Key Vault and pipeline integration.

What is RTO & RPO?

Answer:

- RTO → Recovery Time Objective
- RPO → Recovery Point Objective
- Used in disaster recovery
- Defines downtime & data loss

One-liner:

RTO = recovery time, RPO = acceptable data loss.

What is Network Peering?

Answer :

- Connects VNets
- Private communication
- Low latency
- No internet involved

One-liner:

VNet peering connects networks privately.

What is Azure Availability Set vs Scale Set?

Answer :

- Availability Set → fault isolation
- Scale Set → scaling + management
- Scale Set supports auto scaling
- Availability Set does not

One-liner:

Availability Set ensures uptime, Scale Set ensures scaling.

What happens if NSG and Firewall conflict?

Answer :

- Both rules apply
- Most restrictive rule wins

- Traffic evaluated step by step
- Logging helps debugging

One-liner:

The most restrictive rule is applied in case of conflict

What is your deployment strategy preference?

Answer :

- Blue-Green → safer
- Rolling → gradual
- Canary → controlled release
- Depends on use case

One-liner:

Choose deployment strategy based on risk and application type.

How do you ensure compliance in Azure?

Answer :

- Use Azure Policy
- Audit logs
- Role-based access
- Security Center
- Regular reviews

One-liner:

Compliance is ensured using policies, auditing, and access control.

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- Monitor via cost management

One-liner:

Cost optimization is done using scaling, reservations, and monitoring usage.

How do you connect on-premises to Azure?

Answer :

- Use **VPN Gateway** (quick setup)
- Use **ExpressRoute** (private connection)
- Configure VNet integration
- Secure with NSG & firewall

One-liner:

Hybrid connectivity is achieved via VPN or ExpressRoute.

How do you handle disaster recovery?

Answer :

- Use Azure Site Recovery
- Replicate to another region
- Use backups
- Test failover regularly

- Define RTO & RPO

One-liner:

Disaster recovery ensures business continuity via replication and failover.

Difference: Azure VM vs App Service

Answer :

- VM → Full control
- App Service → Managed platform
- VM → OS level access
- App Service → No OS management

One-liner:

VM gives full control, App Service is fully managed

What is fault domain vs update domain?

Answer :

- Fault domain → Hardware failure isolation
- Update domain → Maintenance isolation
- Used in Availability Set
- Prevents downtime

One-liner:

Fault domain = hardware isolation, update domain = maintenance isolation.

How do you reduce latency globally?

Answer :

- Use Azure Front Door
- Use **CDN for static content**
- Deploy apps in multiple regions
- Use Traffic Manager

One-liner:

Global latency is reduced using CDN and global load balancing.

How do you implement Zero Downtime Deployment?

Answer :

- Use deployment slots (App Service)
- Blue-Green deployment
- Rolling updates
- Health checks before swap

One-liner:

Zero downtime is achieved using blue-green or rolling deployments.

How do you enforce governance in Azure?

Answer :

- Use **Azure Policy**
- Use **Blueprints**
- Tagging strategy
- Role-based access

One-liner:

Governance is enforced using policies, roles, and standards.

Hub-Spoke Architecture in Azure?

Answer :

- Hub → central services (firewall, VPN)
- Spoke → application VNets
- Improves security
- Scalable design

One-liner:

Hub-Spoke centralizes networking and improves security.

What is Azure Data Factory?

Answer :

- ETL/ELT service
- Data integration
- Supports pipelines
- Connects multiple sources

One-liner:

Data Factory is used for data integration and transformation.

How do you choose between Azure services?

Answer :

- Based on workload
- Cost vs performance
- Scalability needs
- Management overhead
- Security requirements

One-liner:

Service selection depends on workload, cost, and scalability needs.

What is the biggest challenge in cloud?

Answer :

- Cost management
- Security
- Complexity
- Skill gap
- Governance

One-liner:

Managing cost, security, and complexity is the biggest challenge.

What is your approach to troubleshooting in Azure?

Answer :

- Check logs (Monitor)
- Validate configuration
- Check network rules
- Use diagnostics
- Reproduce issue

One-liner:

Troubleshooting involves logs, configuration checks, and diagnostics.

How do you handle production issues?

Answer :

- Identify root cause
- Check monitoring alerts
- Apply quick fix
- Communicate with team

- Prevent future issues

One-liner:

Handle production issues by fixing fast and preventing recurrence.

How do you design a secure network architecture in Azure?

Answer :

- Use VNet with subnets (segmentation)
- Apply **NSG rules**
- Use **Azure Firewall** for centralized control
- Implement **Private Endpoints**
- Enable DDoS protection

One-liner:

Secure architecture uses network isolation, firewalls, and private access.

. Difference: Azure Front Door vs Traffic Manager

Answer :

- Front Door → Layer 7 (HTTP/HTTPS)
- Traffic Manager → DNS-based
- Front Door → real-time routing
- Traffic Manager → DNS resolution-based

One-liner:

Front Door is L7 routing, Traffic Manager is DNS-based routing.

How do you handle secrets in CI/CD?

Answer:

- Use **Azure Key Vault**
- Integrate with pipeline
- Avoid hardcoding
- Use Managed Identity
- Rotate secrets

One-liner:

Secrets are managed securely using Key Vault and pipeline integration.

What is RTO & RPO?

Answer:

- RTO → Recovery Time Objective
- RPO → Recovery Point Objective
- Used in disaster recovery
- Defines downtime & data loss

One-liner:

RTO = recovery time, RPO = acceptable data loss.

What is Network Peering?

Answer :

- Connects VNets
- Private communication
- Low latency
- No internet involved

One-liner:

VNet peering connects networks privately.

What is Azure Availability Set vs Scale Set?

Answer :

- Availability Set → fault isolation
- Scale Set → scaling + management
- Scale Set supports auto scaling
- Availability Set does not

One-liner:

Availability Set ensures uptime, Scale Set ensures scaling.

What happens if NSG and Firewall conflict?

Answer :

- Both rules apply
- Most restrictive rule wins
- Traffic evaluated step by step
- Logging helps debugging

One-liner:

The most restrictive rule is applied in case of conflict

What is your deployment strategy preference?

Answer :

- Blue-Green → safer
- Rolling → gradual
- Canary → controlled release
- Depends on use case

One-liner:

Choose deployment strategy based on risk and application type.

How do you ensure compliance in Azure?

Answer :

- Use Azure Policy
- Audit logs
- Role-based access
- Security Center
- Regular reviews

One-liner:

Compliance is ensured using policies, auditing, and access control.

